

# ACM-1

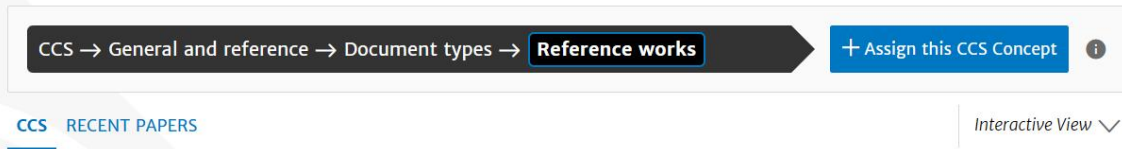
| ACM-为提高文章通过率，请作者严格按照以下标准自检文章版面 |                                                                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 页面设置                           | 页边距：上：3.1 厘米，下：5.01 厘米，左：2.54 厘米，右：3.6 厘米<br>纸张大小：Letter，宽度：21.59 厘米，高度：27.94 厘米<br>无网格                                                                                                            |
| 篇幅                             | 全文单栏不能低于 8 页，双栏不能低于 5 页                                                                                                                                                                            |
| 作者                             | 请在文中备注所有作者的邮箱，每位作者的邮箱均是唯一的（通讯作者的邮箱必须用符号“*”标记清楚且邮箱必须有效，以便接收签署论文出版版权的邮件。）                                                                                                                            |
| CCS                            | 根据 ACM 的出版要求，请在正文中关键词的下面补充 CCS CONCEPTS 部分。CCS 操作如下：登录 <a href="https://dl.acm.org/ccs">https://dl.acm.org/ccs</a> 选择,根据自己研究方向选择类目，复制张贴 CCS 中所选类目的文字。 <b>并将 CCS 的代码复制在文件的备注栏（请参考 ACM-2 的操作指引）。</b> |
| 字体                             | 全文字体：Linux Biolinum O；字号：小五；两端对齐（除专有名词、所有一级标题、以及首句首首字母字母大写以外，其余单词全部小写）                                                                                                                             |
| 标题                             | Paper Title (字号小四，加粗，左对齐)                                                                                                                                                                          |
|                                | 一级标题格式统一（ <b>大写</b> ，字号 小五，加粗）                                                                                                                                                                     |
|                                | 二级标题格式统一（字号 小五，加粗）                                                                                                                                                                                 |
|                                | 三级标题格式统一（字号 小五，斜体，加粗）                                                                                                                                                                              |
| 图片                             | 图：图片必须要高清，且图片中不能有中文字符。                                                                                                                                                                             |
|                                | 图：每张图片仅对应一个标题，如果有多张照片，请将多张照片整合成一张。                                                                                                                                                                 |
|                                | 图片标题：字号 8，居中（所有“Figure”不能缩写，标题末尾需要加句号）。所有图片必须要在正文中引用，且按阿拉伯数字顺序排序（如 Figure 1、2、3.....）                                                                                                              |
| 表格                             | 表格标题：字号 8，居中（所有“Table”不能缩写，标题末尾需要加句号）                                                                                                                                                              |
|                                | 所有表格必须要在正文中引用，且按阿拉伯数字顺序排序（如 Table 1、2、3.....）                                                                                                                                                      |
| 公式                             | 所有公式均需要可编辑格式，且阿拉伯按顺序排序。（如：(1)、(2)、(3).....）                                                                                                                                                        |
| 文献                             | 所有文献必须注明出版年份                                                                                                                                                                                       |
|                                | 参考文献不能低于 5 条，且被引用的文献作者来源应来自三个或者三个以上国家，参考文献在正文中的引用需要以“ <sup>[阿拉伯数字]</sup> ”上标的形式标注。                                                                                                                 |

**PS：请联系所有共同作者提供唯一的 ORCID 号备用，后续签署出版协议的时候需要填写此信息。如果没有，可提前访问此链接创建：<https://orcid.org/register>**

\*由于 ACM 的排版要求比复杂，所以作者只要确保以上需要注意的地方均按照要求核实无误即可，文章交付出版前，排字编辑将会按照出版物的要求进行排版，并会联系作者确认终稿。

CCS 张贴指引：

- 1、访问 CCS 链接：<https://dl.acm.org/ccs>
- 2、根据文章的研究方向，选择符合文章内容相关主题的（三级）类目



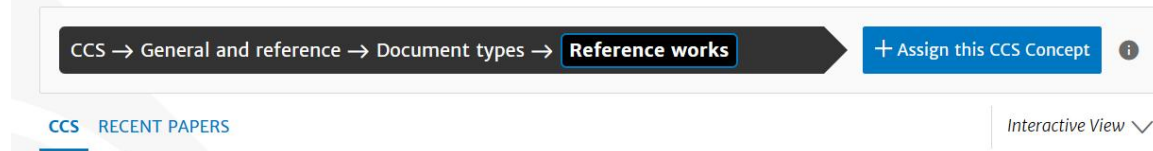
- 3、张贴 CCS（复制即可，不要进行任何更改）。

CCS CONCEPTS

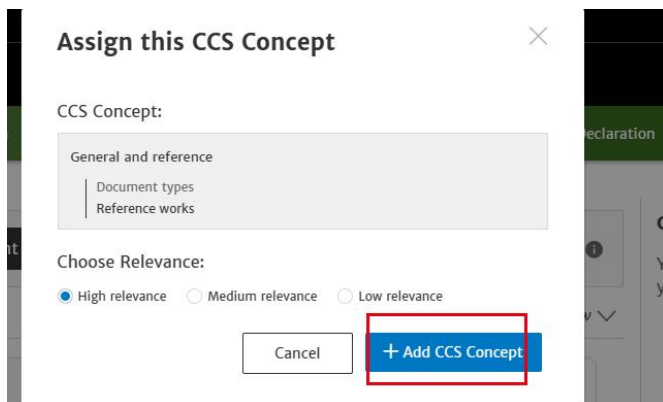
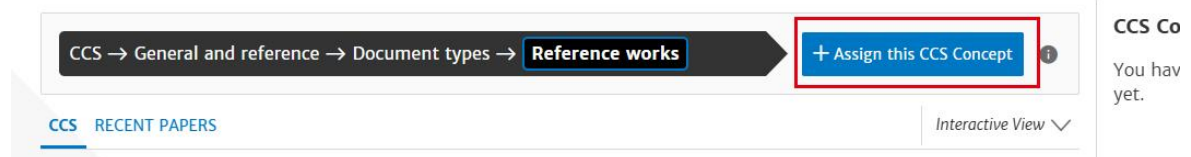
General and reference ~ Document types ~ Document types

- 4、复制代码

① 选择类目



② 添加 CCS



## ③ 创建代码

The screenshot shows the CCS Concept interface. At the top, there's a breadcrumb trail: CCS → General and reference → Document types → **Reference works**. A button '+ Assign this CCS Concept' is visible. Below this, there's a grid of document types: Surveys and overviews, Reference works, General conference proceedings, Biographies, General literature, and Computing standards, RFCs and guidelines. On the right, the 'CCS Concept' panel shows 'General and reference' as the selected category, with 'Reference works' as the document type. A red box highlights the '<> Generate Code' button. Below this, a modal dialog titled 'Generate Code' is open. It contains instructions: 'The code below should be used with a TeX based typesetting system. Copy the code using the icon below and paste it into the item you are classifying only if you have fully classified the item.' There's a checkbox 'Show the XML only' which is checked, and a red box around it with the text '一定要勾选' (Must be checked). Below the checkbox is a code block containing XML for <ccs2012>. A red box highlights the copy icon at the bottom right of the code block.

CCS Concept

Here you can view and edit your current CCS concept or [clear all](#) your previous selections.

General and reference

High Relevance

Document types

Reference works

**<> Generate Code**

[Preview CCS Chart](#)

**Generate Code**

The code below should be used with a TeX based typesetting system. Copy the code using the icon below and paste it into the item you are classifying only if you have fully classified the item.

☒ Show the XML only **一定要勾选**

```
<ccs2012>
  <concept>
    <concept_id>10002944.10011122.10002946</concept_id>
    <concept_desc>General and reference~Reference works</concept_desc>
    <concept_significance>500</concept_significance>
  </concept>
</ccs2012>
```

## ④ 复制代码至备注栏。备注路径：“文件”→“信息”→“备注”。

The screenshot shows the 'Template ACM-2.0' document interface. On the left, there's a sidebar with options: 开始 (Start), 新建 (New), 打开 (Open), 获取加载项 (Get Add-ins), 信息 (Information), 保存 (Save), 另存为 (Save As), and 保存到百度网盘 (Save to Baidu Netdisk). The '信息' (Information) tab is selected. The main area shows document details: 'Template ACM-2.0', 'H:\硬盘文件合集\个人文件\出版物模板\ACM-出版物模板', and buttons for 上传 (Upload), 共享 (Share), 复制路径 (Copy Path), 复制本地路径 (Copy Local Path), and 打开文件位置 (Open File Location). Below this, there are sections for '保护文档' (Protect Document) and '检查文档' (Check Document). On the right, there's a '属性' (Properties) panel showing details like 大小 (Size: 328KB), 页数 (Pages: 17), 字数 (Word Count: 6478), and 编辑时间总计 (Total Editing Time: 226 minutes). At the bottom, there's a '备注' (Remarks) field with a red box around it, containing the XML code: <begin>(CCSXML)<br><ccs2012><br><concept> ...

信息

Template ACM-2.0

H:\硬盘文件合集\个人文件\出版物模板\ACM-出版物模板

上传 共享 复制路径 复制本地路径 打开文件位置

**保护文档**

控制其他人可以对此文档所做的更改类型。

**检查文档**

在发布此文件之前，请注意是否包含：

- 文档属性、作者的姓名和视图图像数据
- 页眉和页脚
- 以隐藏文字为格式的文字

属性

大小 328KB

页数 17

字数 6478

编辑时间总计 226 分钟

标题 添加标题

标记 添加标记

**备注** <begin>(CCSXML)<br><ccs2012><br><concept> ...

## Submission Template for ACM Papers

First Author's Name \* (此标记一定是通讯作者的邮箱，不可进行任何更改，要确保能接收签署版权的邮件)

First author's affiliation, an Institution with a very long name, city, post number, Country, xxxx@gmail.com

Second Author's Name

First author's affiliation, an Institution with a very long name, city, post number, Country, xvxx@gmail.com

Third Author's Name

First author's affiliation, an Institution with a very long name, city, post number, Country, xaaaxx@gmail.com

### Abstract

Although there is no distinctive header, this is the abstract. This submission template allows authors to submit their papers for review to an ACM Conference or Journal without any output design specifications incorporated at this point in the process. The ACM manuscript template is a single column document that allows authors to type their content into the pre-existing set of paragraph formatting styles applied to the sample placeholder text here. Throughout the document you will find further instructions on how to format your text. If your conference's review process will be double-blind: The submitted document should not include author information and should not include acknowledgments, citations or discussion of related work that would make the authorship apparent. Submissions containing author identifying information may be subject to rejection without review. Upon acceptance, the author and affiliation information must be added to your paper.

### CCS CONCEPTS

General and reference ~ Document types ~ Document types

### Keywords

Keyword number 1, Keyword number 2, Keyword number 3, Keyword number 4

## 1 INTRODUCTION

ACM's new manuscript submission template aims to provide consistent styles for use across ACM publications, and incorporates accessibility and metadata-extraction functionality necessary for future Digital Library endeavors. Numerous ACM and SIG-specific templates have been examined, and their unique features incorporated into this single new template. If you are new to publishing with ACM, this document is a valuable guide to the process of preparing your work for publication. If you have published with ACM before, this document provides insight and instruction into the current process for preparing your manuscript.

### 1.1 Accessibility

Following the guidelines throughout this template will also improve the accessibility of your manuscript and increase the audience for your work. Ensure that heading styles are applied as instructed, tables are created using Word's table feature (rather than an image), figures have a text equivalent, and list styles are applied as instructed.

### 1.1.2 More about the submission template. (一定要句号)

This submission version of your paper should not have headers or footers, these will be added when your manuscript is processed after acceptance. It should remain in a one-column format—please do not alter any of the styles or margins.

## 2 INSERTING CONTENT ELEMENTS

The next subsections provide instructions on how to insert figures, tables, and equations in your document.

### 2.1 Tables

Tables are “float elements” which should be inserted after their first text reference and have specific styles for identification. Do not use images to present tables, or they will be inaccessible to readers using assistive technologies.

Table 1. Styles available in the Word template.

| Style Tag           | Definition                                                          | Style Tag     | Definition                   |
|---------------------|---------------------------------------------------------------------|---------------|------------------------------|
| Title_document      | main title of article                                               | ListParagraph | list items                   |
| Subtitle            | subtitle of article                                                 | Statements    | math statements              |
| Authors             | author name                                                         | Extract       | block quotations             |
| Affiliation         | author affiliation information                                      | Algorithm     | caption for algorithm        |
|                     |                                                                     | Caption       |                              |
| AuthNotes           | footnote to author(s)                                               | AckHead       | heading for acknowledgements |
| Abstract            | abstract text                                                       | AckPara       | acknowledgements text        |
| CCSHead             | heading for CSS Concepts                                            | GrantSponsor  | sponsor of grant             |
| CCSDescription      | CSS terms                                                           | GrantNumber   | number for the grant         |
| KeyWordHead         | heading for keywords                                                | ReferenceHead | heading for references       |
| Keywords            | keywords text                                                       | Bib_entry     | references                   |
| ORCID               | author's ORCID #                                                    | AppendixH1    | appendix heading level 1     |
| Head1               | heading level 1                                                     | AppendixH2    | appendix heading level 2     |
| Head2               | heading level 2                                                     | AppendixH3    | appendix heading level 3     |
| Head3               | heading level 3                                                     | TableCaption  | title of table               |
| PostHeadPara        | first paragraph after a heading                                     | TableHead     | column head of table         |
|                     |                                                                     | TableFootnote | footnote to table            |
| Para                | Subsequent paragraphs of general text                               | Image         | figures                      |
| ParaContinue        | flush left text after display items like math equations, lists etc. | DOI           | Digital object identifier    |
| DisplayFormula      | numbered math equation                                              |               |                              |
| DisplayFormulaUnnum | unnumbered equations                                                | Label         | label <sup>a</sup>           |
| ComputerCode        | Display Computer codes                                              | In-text code  | intext computer code         |
| Short Title         | Short title of article                                              |               |                              |

### 2.2 Figures

Figures are “float elements” which should be inserted after their first text reference, and have specific styles for identification. Insert a figure and apply the “**Image**” paragraph style to it. For the figure caption, apply the style “**FigureCaption.**”



Figure 1. 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons.  
(<https://goo.gl/VLCRBB>)



Figure 3. The layout of multipart images should be as per the above example within the table. All images must have the “Image” style applied.

## 2.3 Equations

There are two types of math equations: the *numbered display math equation* and the *un-numbered display math equation*. Below are examples of both.

### 2.3.1 DisplayFormula.

The **DisplayFormula** style is applied in the numbered math equation. A numbered display equation always has an equation number (label) on the right.

$$\frac{x = -b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

## 2.4 Algorithms (算法不要放在表格中)

Algorithms use the styles “AlgorithmCaption” and “Algorithm”.

---

### ALGORITHM 1: Iterative Algorithm

---

```
current_position    center
current_direction   up
current_position is inside circle
while current_position is inside circle, do
    neighborhood all grid hexes within two hexes from current_position
    for each hex in neighborhood, do
        for each neuron in hex do
            convert neuron_orientation to vector
            scale vector by neuron_excitation
            vector_sum vector_sum + vector
        end
    end
end
normalize vector_sum
end
```

---

## 3 CONCLUSION

Acknowledgments are placed before the references. Add information about grants, awards, or other types of funding that you have received to support your research. Author can capture the grant sponsor information, by selecting the grant sponsor text and apply style ‘GrantSponsor’. After this, select grant no and apply ‘GrantNumber’ from style panel. Example of Grant sponsor: Competitive Research Programme and example of Grant no: CRP 10-2012-03.

## ACKNOWLEDGMENTS

Acknowledgments are placed before the references. Add information about grants, awards, or other types of funding that you have received to support your research. Author can capture the **grant sponsor information**, by selecting the grant sponsor text and apply style ‘GrantSponsor’. After this, select grant no and apply ‘GrantNumber’ from style panel. Example of Grant sponsor: [Competitive Research Programme](#) and example of Grant no: [CRP 10-2012-03](#).

## REFERENCES (一定要有出版年份)

- [1] Atul Adya, Paramvir Bahl, Jitendra Padhye, Alec Wolman, and Lidong Zhou. **2004**. A multi-radio unification protocol for IEEE 802.11 wireless networks. In Proceedings of the IEEE 1st International Conference on Broadnets Networks (BroadNets’04) IEEE, Los Alamitos, CA, 210–217. <https://doi.org/10.1109/BROADNETS.2004.8>.
- [2] Sam Anzaroot and Andrew McCallum. 2013. UMass Citation Field Extraction Dataset. Retrieved May 27, **2019** from <http://www.iesl.cs.umass.edu/data/data-umasscitationfield>.
- [3] Martin A. Fischler and Robert C. Bolles. **1981**. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. Commun. ACM 24, 6 (June 1981), 381–395.